

Harish Kumar S

AI/ML Engineer

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<https://harish-nika.github.io/>

PROFESSIONAL SUMMARY

Integrated MTech in Data Science (VIT, 2025) AI/ML engineer building production systems at FactEntry Data Solutions (A SIX Company). Shipped the Data Mining Engine on K3s with GitLab CI, Argo CD GitOps, Redis queue workers, and FastAPI microservices. Also delivered RAG document systems, PyPI-published NLP tooling, Roboflow computer-vision models, and containerised AI deployments.

- Production K3s bond-extraction platform (80+ CPU cores, Redis streams, human-in-the-loop validation)
- GitOps: GitLab CI -> Helm promote -> Argo CD sync to stage
- RAG and document intelligence: FAISS, Groq LLaMA, Streamlit/FastAPI
- PyPI package: language-processing-tool | Roboflow YOLOv11 dental CV models

CORE COMPETENCIES

AI/ML Engineer | Machine Learning | Deep Learning | Computer Vision | NLP | RAG | Python | FastAPI | Kubernetes | K3s | Docker | Redis | PostgreSQL | GitLab CI | Argo CD | Helm | GitOps | MLOps | Roboflow | YOLO | FAISS | LangChain | Streamlit

EXPERIENCE

Software Developer Trainee (Probation)

FactEntry Data Solutions - A SIX Company | Dec 2025 - Present

- Shipped Data Mining Engine (DME) modules on K3s stage: corporate/municipal bond extraction, validation UI, and cluster operations portal.
- Implemented Redis-stream job dispatch with multi-node worker throughput and GitOps delivery (GitLab CI image build -> Helm chart promote -> Argo CD).
- Built FastAPI microservices, live WebSocket job telemetry, role-based CPU tiering, and domain-restricted enterprise auth.

MLOps and Data Science Intern

FactEntry Data Solutions - A SIX Company | Nov 2024 - May 2025

- Published language-processing-tool on PyPI (PDF language detection, OCR, document clustering) as self-driven internship contribution.
- Built Fixed Income Expert RAG system (FAISS, Groq LLaMA 3, Streamlit/FastAPI) as personal contribution during internship.

Associate Software Engineer Trainee Intern

MIMASOFT Technologies | Jun 2025 - Oct 2025

- Developed Dental ML Project: Roboflow YOLOv11 tooth and panoramic X-ray models with inference CLI.
- Contributed to backend integration, API workflows, and production-oriented software development practices.

EDUCATION

Integrated MTech - CSE (Data Science), Vellore Institute of Technology (2020 - 2025) - CGPA 7.8

Class XII, THIRU G V C Higher Secondary School (2020) - 79%

Class X, St Joseph Matriculation Higher Secondary School (2018) - 87.8%

TECHNICAL SKILLS

AI/ML: Machine Learning, Deep Learning, Generative AI, Computer Vision, NLP, RAG, FAISS, Sentence Transformers, Ollama, Roboflow, YOLOv11, Supervision

Engineering: Python, FastAPI, Streamlit, SQL, Docker, Kubernetes (K3s), Redis, PostgreSQL, LangChain, PyMuPDF, OpenCV

DevOps/GitOps: GitHub, GitLab CI, Argo CD, Helm, systemd, GitHub Actions, GitOps

PROJECTS

Data Mining Engine (DME)

Where: FactEntry Data Solutions (A SIX Company) - Software Developer Trainee, probation
Enterprise bond extraction on K3s - UI, API, Redis queue workers, GitOps CI/CD, validation workspace.
Tech: K3s, FastAPI, Redis, PostgreSQL, Argo CD
<https://harish-nika.github.io/dme-project.html>

Fixed Income Expert System

Where: FactEntry internship - personal contribution
RAG chatbot for financial bond PDFs - upload, embed, retrieve, LLaMA 3 answers via Groq.
Tech: FastAPI, Streamlit, FAISS, Groq, SBERT
<https://harish-nika.github.io/fexpert.html>
<https://github.com/Harish-nika/FExpert>

Cybersecurity Content Moderator

Where: VIT - Capstone project (academic)
Multimodal hate-speech and harmful content classification (text, image, PDF).
Tech: LLM, Streamlit, Docker, FastAPI
<https://harish-nika.github.io/content-moderator.html>
<https://github.com/Harish-nika/Content-moderator-image>

Language Distribution & Document Clustering Tool

Where: FactEntry internship - self contribution
PyPI-published PDF language detector, OCR, and clustering pipeline (v0.2.7).
Tech: Python, Tesseract, LangDetect, PyPI
<https://harish-nika.github.io/lang-tool.html>
https://github.com/Harish-nika/language_processing_tool

MedBot (Medical RAG Assistant)

Where: VIT - J Component, Soft Computing course
Medical PDF Q&A with SBERT + FAISS retrieval and Groq LLaMA 3 grounded responses.
Tech: Streamlit, FAISS, Sentence Transformers, Groq
<https://harish-nika.github.io/medbot.html>
<https://github.com/Harish-nika/medbot>

Dental ML Project

Where: MIMASOFT Technologies - trainee internship
YOLOv11 tooth and panoramic X-ray detection - Roboflow fine-tuning and inference CLI.
Tech: Roboflow, YOLOv11, inference-sdk, Supervision
<https://harish-nika.github.io/dental-project.html>
https://github.com/Harish-nika/Dental_ML_project

Disease Prediction using Ensemble Learning

Where: SmartInternz / Smart Bridge - Applied Data Science externship
Ensemble ML model for clinical trial disease prediction with MongoDB Atlas pipeline.
Tech: Python, Random Forest, SVM, MongoDB
<mailto:harishkumar56278@gmail.com?subject=Disease%20Prediction%20Project>

Real-Time Room Temperature & Humidity Visualization (AR)

Where: VIT - J Component, ARVR course
MQTT telemetry with Unity AR overlay and ML-driven climate insights.
Tech: Unity, MQTT, AR, Python
<mailto:harishkumar56278@gmail.com?subject=AR%20Visualization%20Project>

Extractive Summarization & Information Retrieval

Where: JP Infotech
NLP web scraping with English-to-SQL query translation and 95% extraction accuracy.

Tech: Python, NLP, SQLite, MySQL

mailto:harishkumar56278@gmail.com?subject=Summarization%20Project

Application Tracking System

Where: JP Infotech

Resume analyser with OCR, keyword suggestions, and n-gram typing assist vs job descriptions.

Tech: Python, PyTesseract, OpenCV, NLP

mailto:harishkumar56278@gmail.com?subject=Application%20Tracking%20System

CERTIFICATIONS

- Microsoft Certified: Azure AI Fundamentals

https://www.credly.com/badges/da3ef4b3-03a0-4560-95b6-5941473704de/public_url

- Applied Data Science (SmartInternz + Google Developers)

https://drive.google.com/file/d/126d8VTJ6y-X-t6_S_KOD2gYttWQhqe2T/view

- Data Analysis and Visualization with Power BI

<https://www.coursera.org/account/accomplishments/verify/YKAC25V5T6WJ>

- Preparing Data for Analysis with Microsoft Excel

<https://www.coursera.org/account/accomplishments/verify/F3SR649G78ZN>

- Harnessing the Power of Data with Power BI

<https://www.coursera.org/account/accomplishments/verify/CTOOZYTXL926>

- Extract, Transform, and Load Data in Power BI

<https://www.coursera.org/account/accomplishments/verify/5VXS5AHV12DN>